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St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001

AMT101S - ALLIED MATHEMATICS - I

Time : 3 hrs **Max Marks :75**

SECTION – A (5X2=10)

Answer **ALL** Questions

1. Write the exponential series.
2. State Cayley Hamilton theorem.
3. If $x = r\cos\theta, y = r\sin\theta$ then find $\frac{\partial(x,y)}{\partial(r,\theta)}$.
4. Evaluate $\int x^3 \sin x dx$.
5. Find $L(\cos 3t \cos t)$.

SECTION – B (3X5=15)

Answer any **THREE** Questions

6. Sum the series $1+\frac{1}{3}+\frac{1\cdot 3}{3\cdot 6}+\frac{1\cdot 3\cdot 5}{3\cdot 6\cdot 9}+....$
7. Verify Cayley Hamilton theorem for $\begin{bmatrix} 8 & -8 & 2 \\ 4 & -3 & -2 \\ 3 & -4 & 1 \end{bmatrix}$.

8. If $u = x^2 + y^2 + z^2, v = x + y + z$ and $w = xy + yz + zx$ then find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$.

9. Find the Fourier series for the function $f(x) = 1 + \frac{2x}{\pi}, -\pi \le x \le 0$, $f(x) = 1 - \frac{2x}{\pi}, 0 \le x \le \pi$

and Deduce that $1 + \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + ... = \frac{\pi^2}{8}$.

10. Find $L\left[e^{2t}\cos 3t\right]$.

SECTION – C (5X10=50)

Answer **ALL** Questions

11. a) Find the sum to infinity of the series

$$\frac{7}{72} + \frac{7 \cdot 28}{72 \cdot 96} + \frac{7 \cdot 28 \cdot 49}{72 \cdot 96 \cdot 120} + ...$$

(or)

b) Sum the series $\sum_{n=1}^{\infty} \frac{n^2 + 3}{n + 2} \frac{x^n}{n!}$.

12. a) Find the eigen values and eigen vectors of $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$

(or)

b) Find the eigen values andeigen vectorsof $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$.

13. a) Show that the radius of curvature at any point of the cardiod $r = a(1+\cos\theta)$ is $\frac{4a}{3}\cos\frac{\theta}{2}$ .Deduce that $\frac{\rho^2}{r}$ is a constant.

(or)

b) If $x + y = 1$, then find the nthderivative of $x^n y^n$.

14. a) Express $f(x) = x\sin x$ as a Fourier Series in $0 \le x \le 2\pi$.

(or)

b) Determine the Fourier series expansion of $x + x^2$ in the interval $(-\pi, \pi)$ and hence deduce that the sum of series $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + ...$

15. a) Using Laplace transform, solve $\frac{d^2y}{dx^2} - y = x^2 + x$ given that $y(0) = y'(0) = 0$.

(or)

b) Find $L^{-1}\left[\frac{1}{s^2 + bs + 13}\right]$.

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